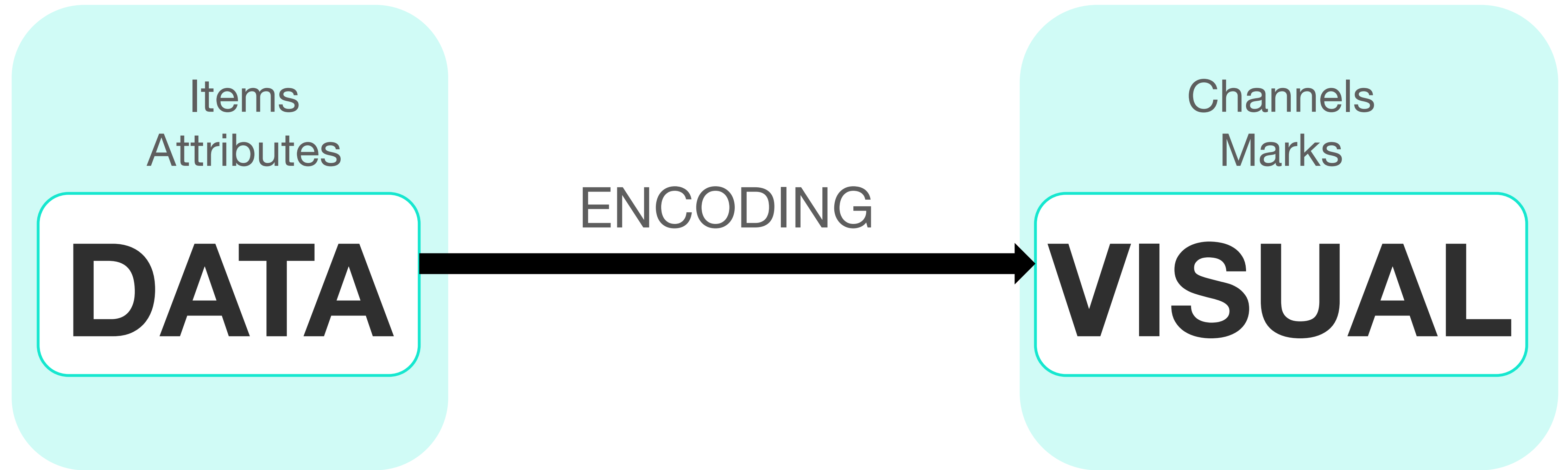


Visual Tasks: From Questions to Sketches

Sarah Groves

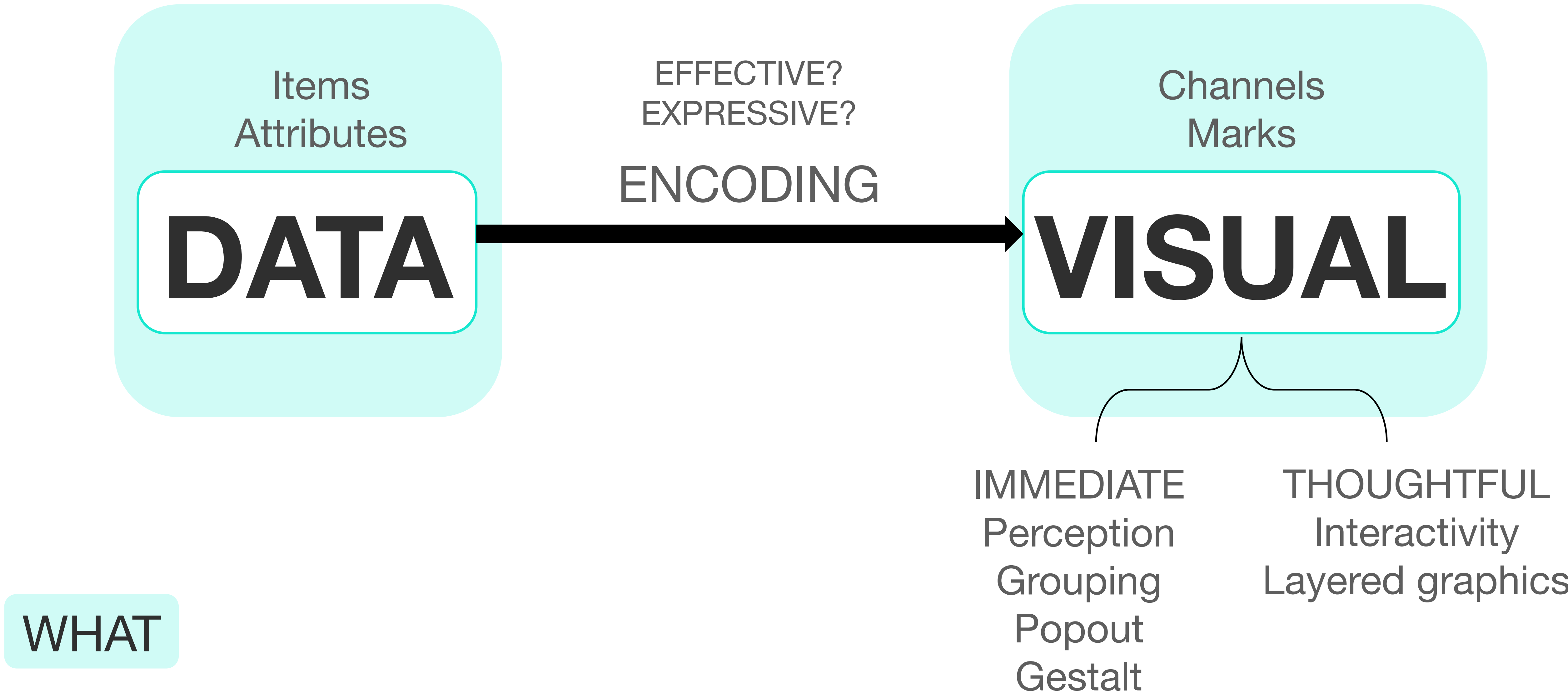
based on materials by Hannah Bako and Zhicheng Liu

How this fits in

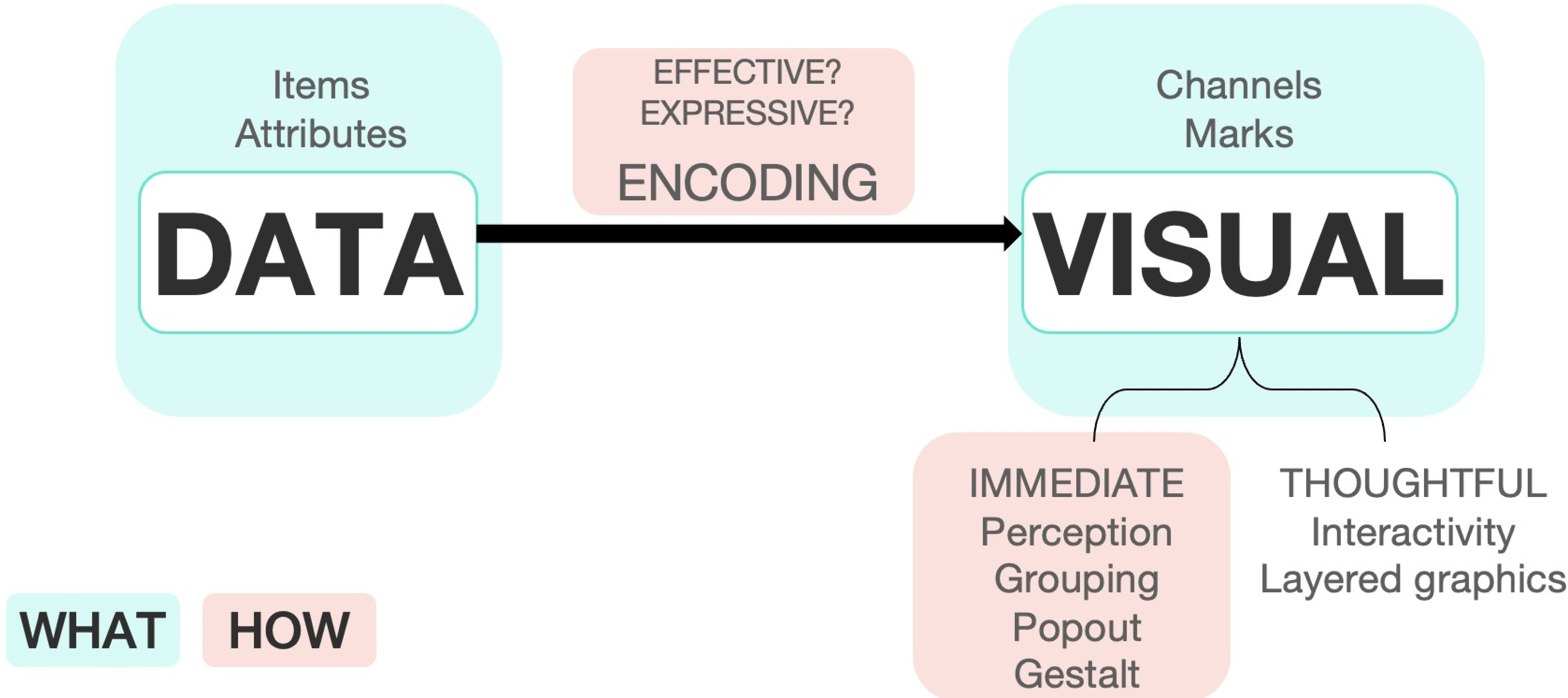


WHAT

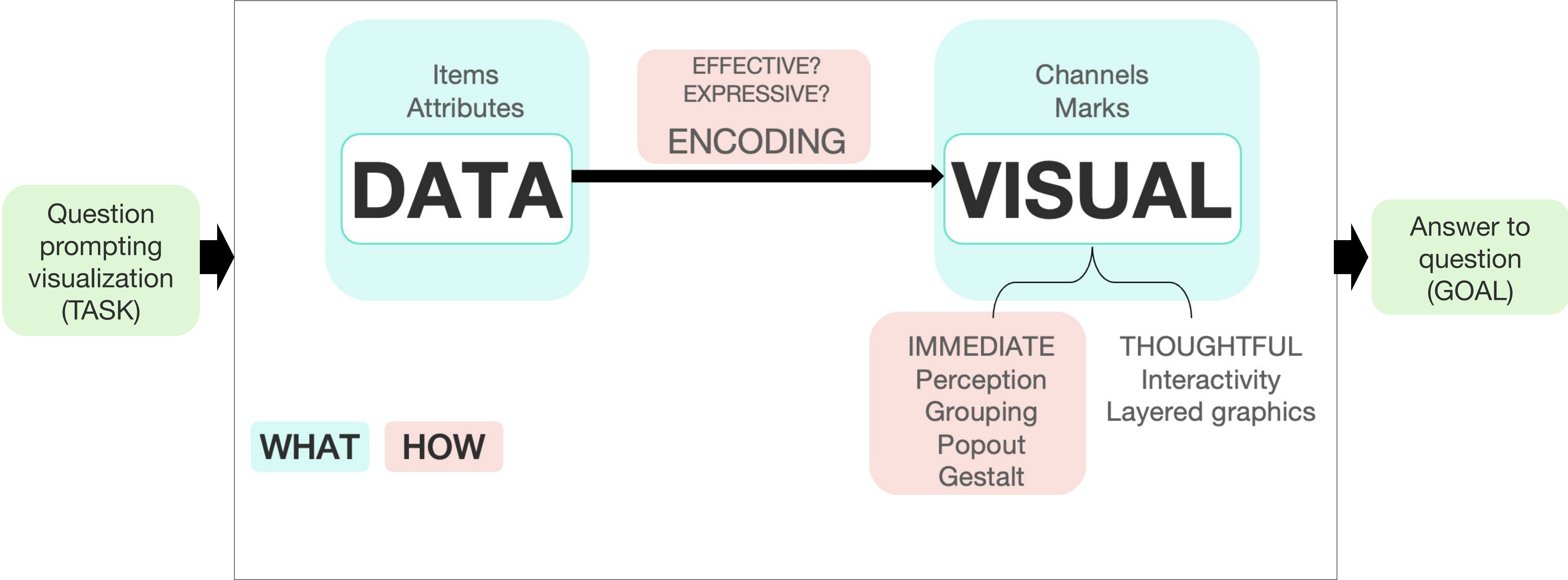
How this fits in



How this fits in



How this fits in

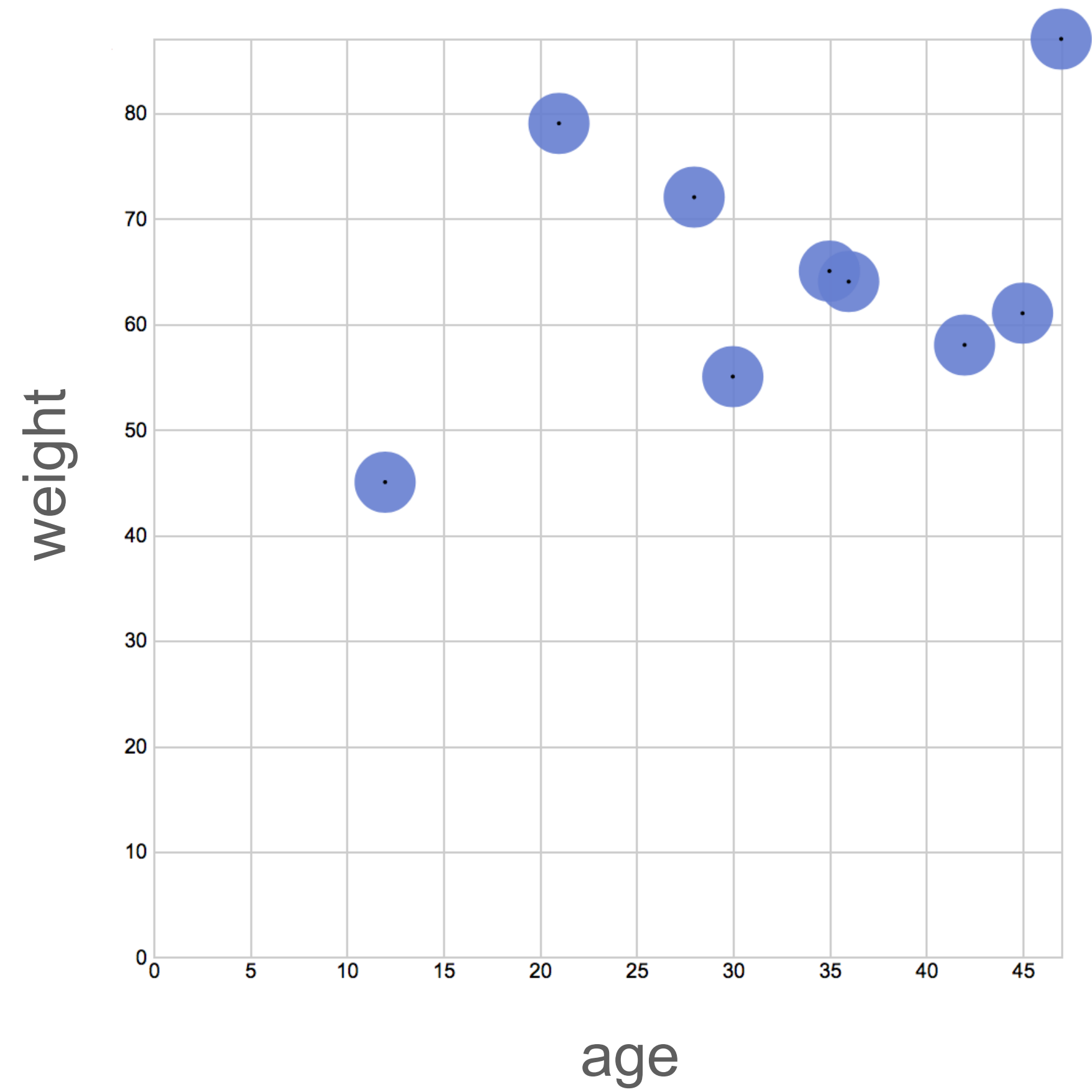


How would you visualize this data table?
Write or sketch a few ideas.

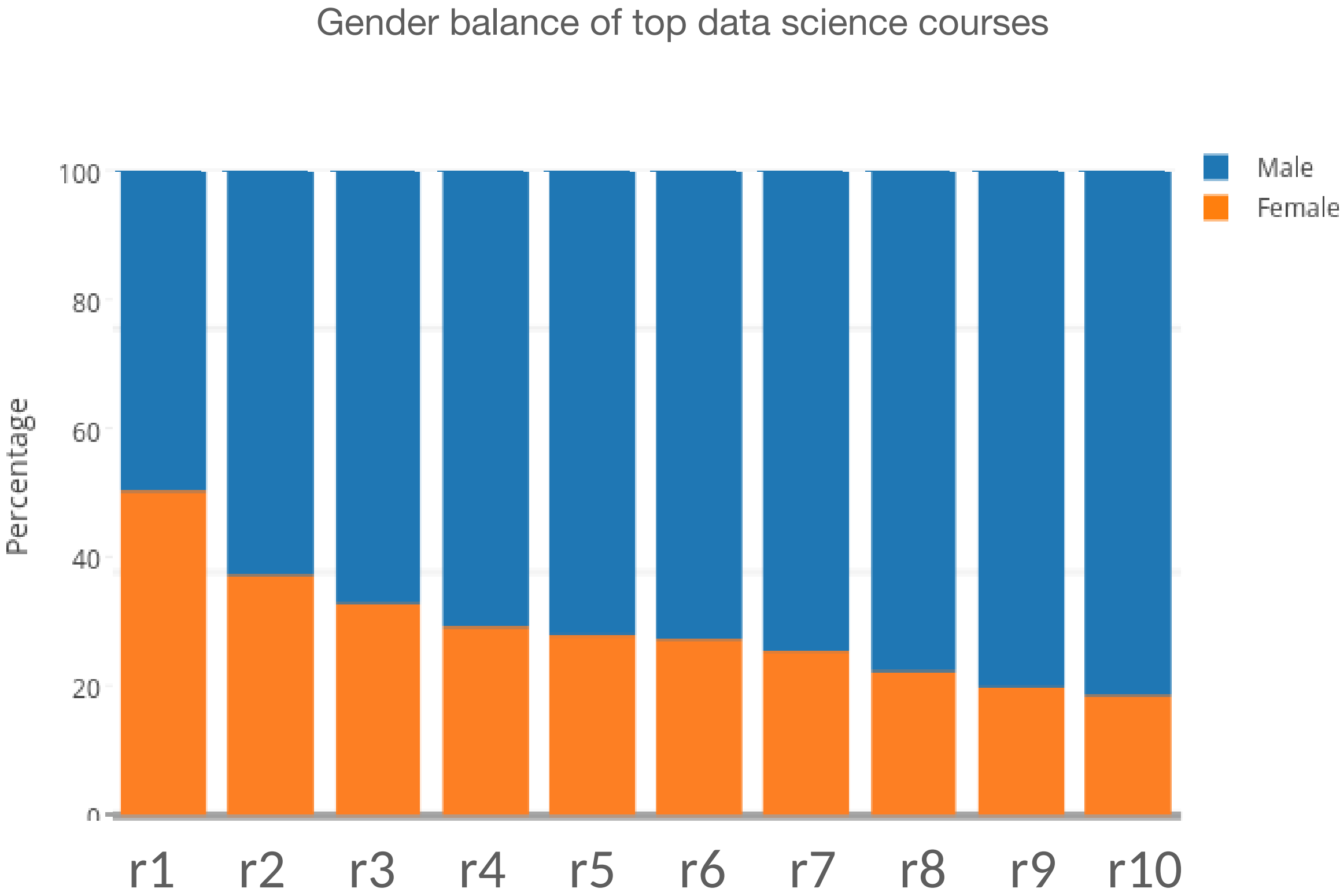
ID	x	y
r1	35	65
r2	21	79
r3	42	58
r4	36	64
r5	28	72
...	...	...

Compare to the person next to you.

ID	x: age	y: weight(kg)
r1	35	65
r2	21	79
r3	42	58
r4	36	64
r5	28	72
...	...	...

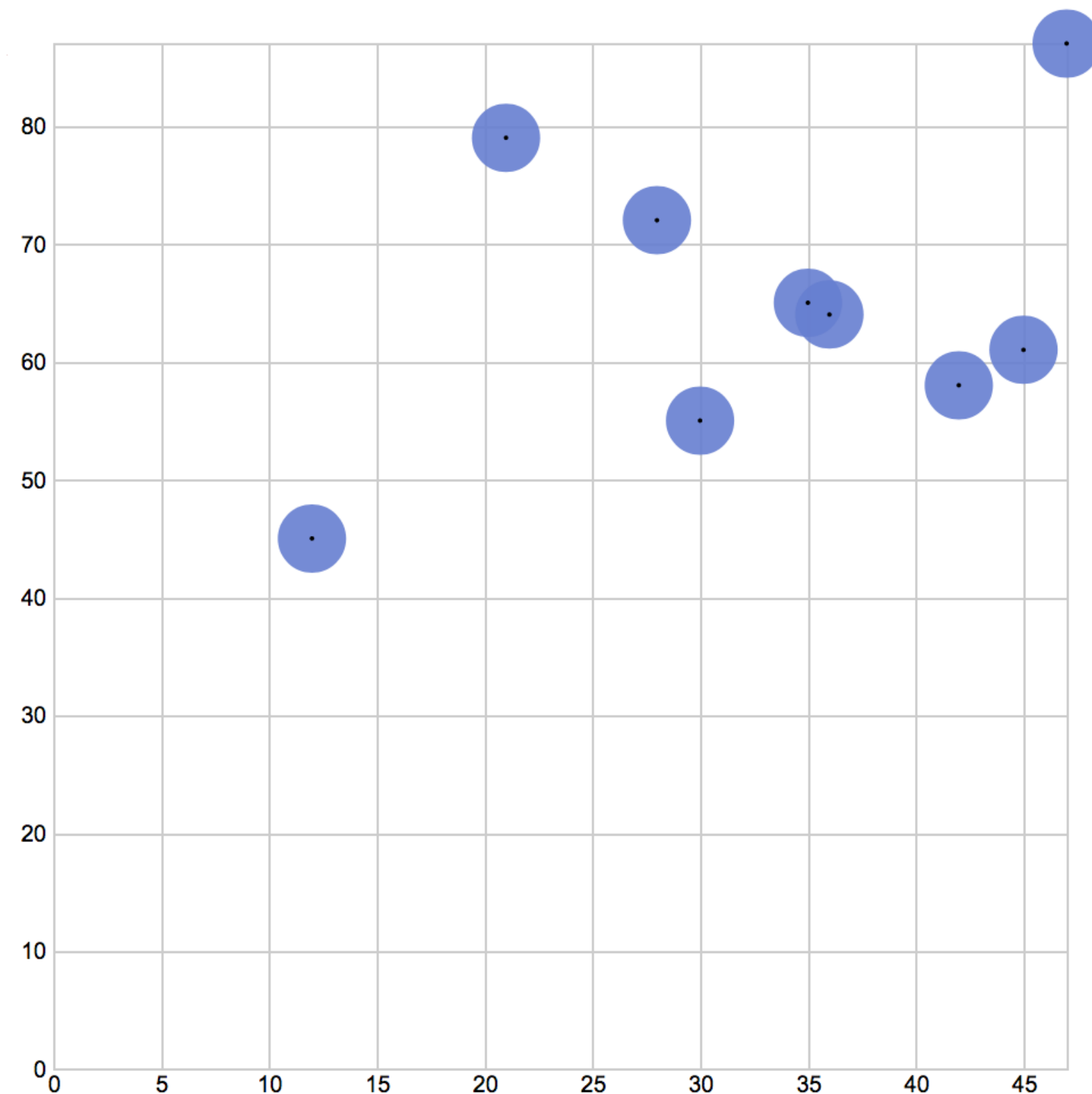


ID	x: female (%)	y: male (%)
r1	35	65
r2	21	79
r3	42	58
r4	36	64
r5	28	72
...	...	...

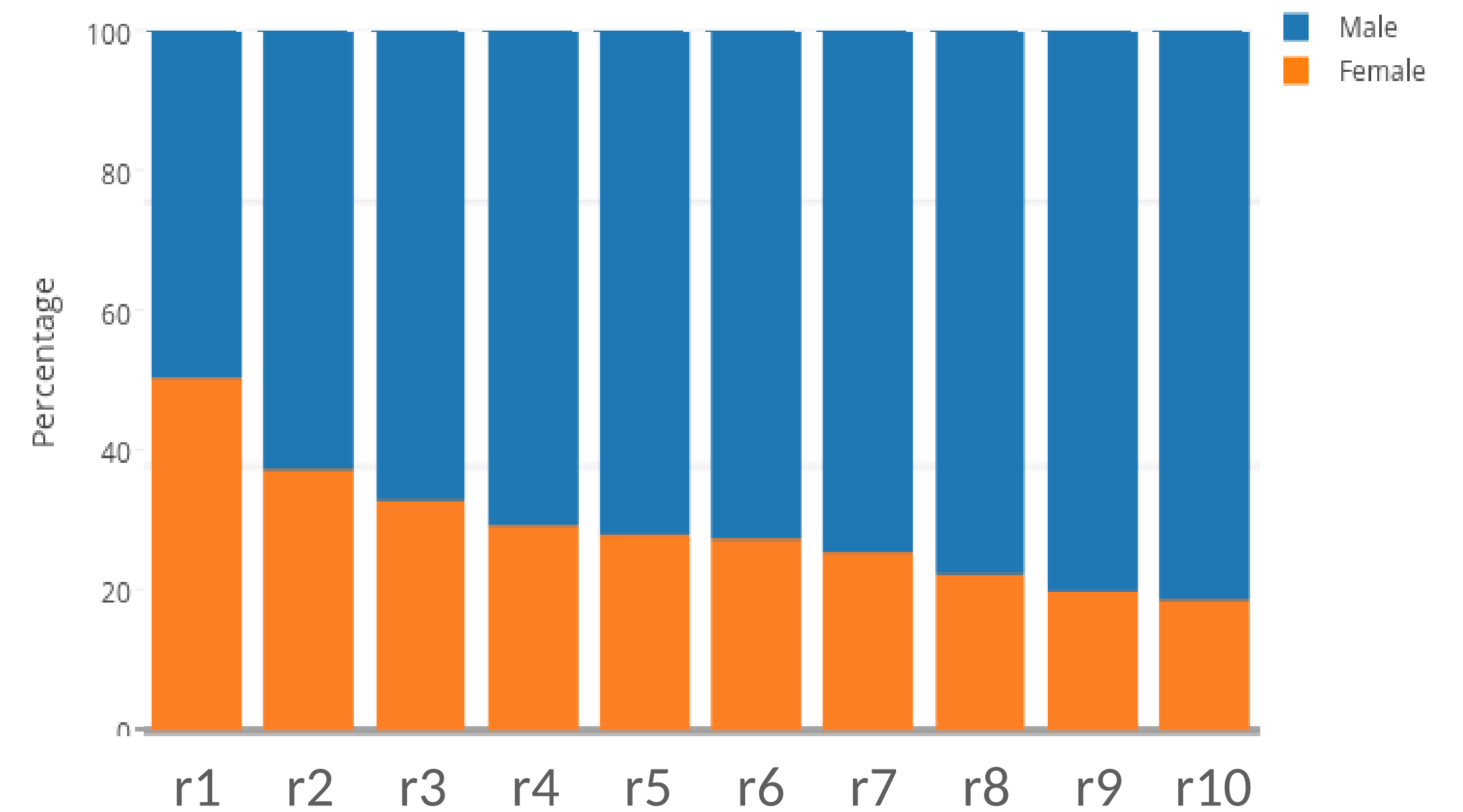


Primary Difference: What the viewer wants to know

Relationship between attributes



Distribution of attribute values



Same data types

But different question =

Different plot

The recipe: question → sketch

1

Choose a question

Choose something specific enough to answer.

2

Name the attributes

Which columns does the question use? What attribute type is each?

3

Decide what one mark is

A row? Or a group summary (count, mean, ratio)? If you have to compute it, that's a derived value.

4

Sketch & check

Pick a form from the guide on the next slide

Check: Can you read the answer in 5 seconds?

The guide: data → questions → plots

You have	You can ask...	Try
Q	What's typical? How spread out? Any outliers?	histogram · box · strip
N	Which is most / least common? How many of each?	sorted bar (counts)
N + Q	Which group is highest? Rank them. Do groups differ in spread?	sorted bar of a summary · box per group
Q + Q	Are they related? Who breaks the pattern? Clusters? How has it changed? When did it peak? Seasonal?	scatter
I/T + Q		line
Grouped plot		
N + N	How do they combine? What share of each group is X?	stacked / 100% bar · count heatmap

L + anything Where is it high / low? map

O ordinal? Treat it like categorical but keep the order on the axis.

Types: N nominal O ordinal Q quantitative I/T time L location

Let's look at life expectancy data

Going through the guide on a specific example

The data

1,704 rows · one row = one country in one year

Columns

With attribute type

- N country: 142 countries
- N continent: 5
- I year: 1952–2007, every 5 yrs
- Q lifeExp: life expectancy (yrs)
- Q pop: population
- Q gdpPercap: GDP per person (\$)
- L iso_alpha: country code (for maps)

year	country	continent	lifeExp	pop	GDP percap
1952	Kenya	Africa	42.3	6.4m	\$854
2007	Kenya	Africa	54.1	35.6m	...
2007	Japan	Asia	82.6	...	...
...	...	...	...	...	...

Question storm

Every question I can think of, tagged by guide row

How many countries in each continent?

N

What is the total GDP per continent?

N + Q (derived)

How has life expectancy changed over time?

O+Q

How has life expectancy changed on each continent?

O+Q+N

Do richer countries live longer?

Q+Q+N

Do increasing populations have longer lives?

O+Q+Q

Where are the countries growing the fastest?

O + Q + L

Are any countries decreasing in life expectancy?

O + N + Q

What is the relationship between GDP and life expectancy?

Q+Q

Is the world getting healthier?

Too vague → refine question

Two moves to get better questions

Sharpen

Make a vague question answerable

“Is the world getting healthier?”



“How has average life expectancy changed from 1952 to 2007 on each continent?”

Stretch

Add “...and does it differ by ____?”



“Do richer countries live longer?”

“...and does that relationship differ by continent?” → **color the dots by continent**

“...and has that changed since 1952?” → **small multiples, one per decade, or *animation***

Which countries grew the most?

Columns

N country

Q $\text{pop 2007} \div \text{pop 1952}$

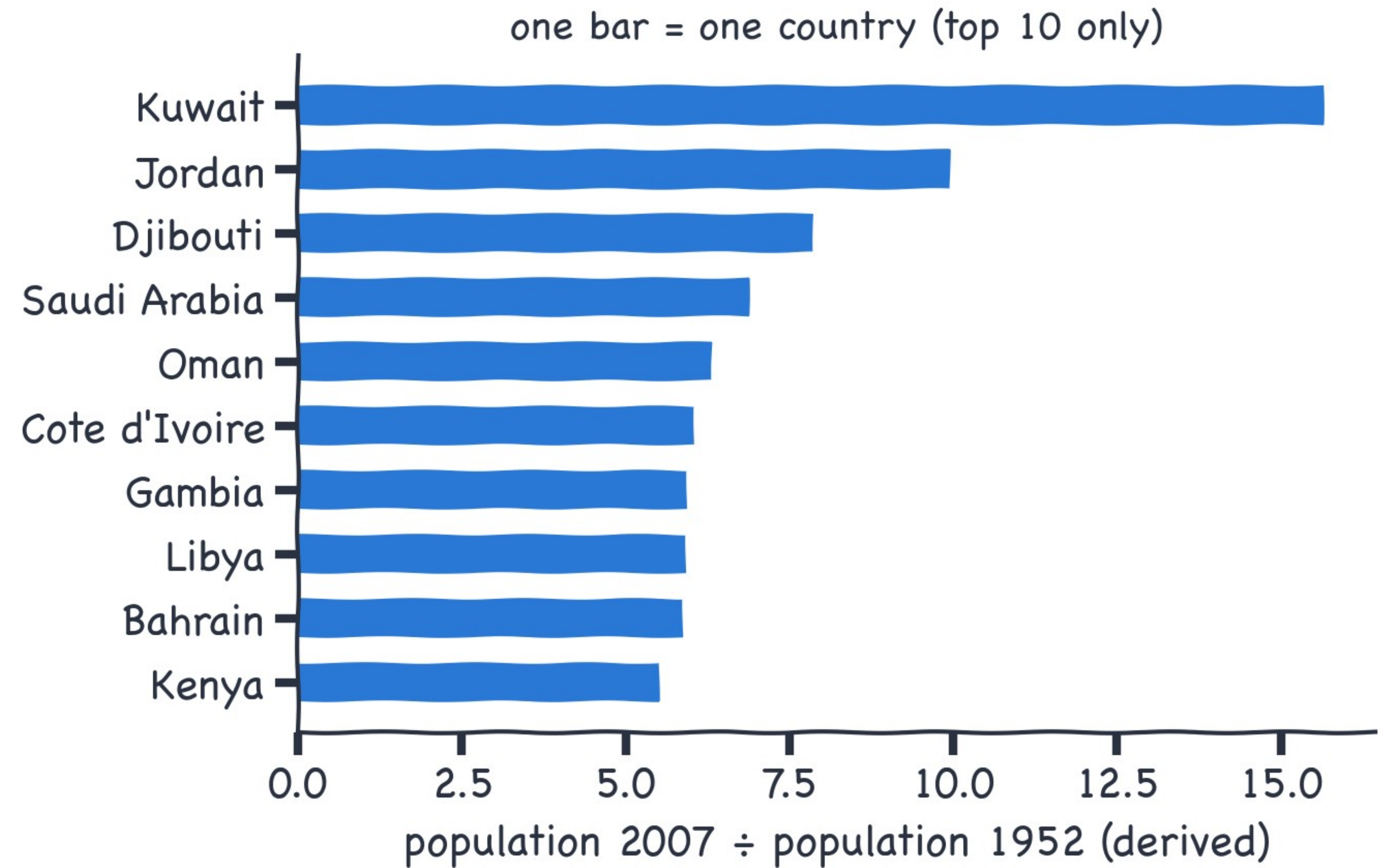
(derived)

One mark =

one country

Form

sorted bar, top 10 only



How has life expectancy changed on each continent?

Columns

I/T year

Q lifeExp

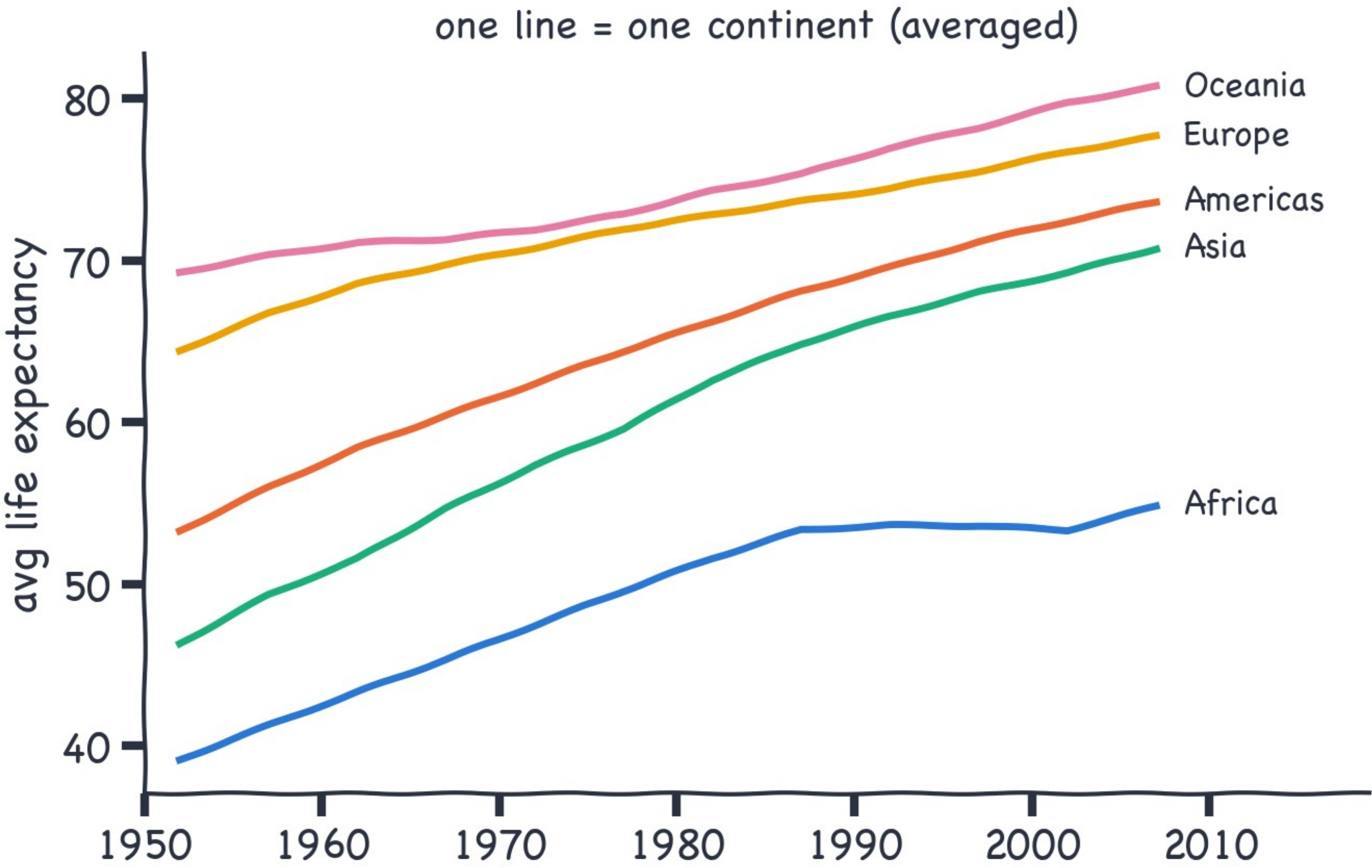
N continent

One mark =

a continent's average in one year

Form

line per continent



Do richer countries live longer?

Columns

Q gdpPercap

Q lifeExp

+ N continent, Q pop

One mark =

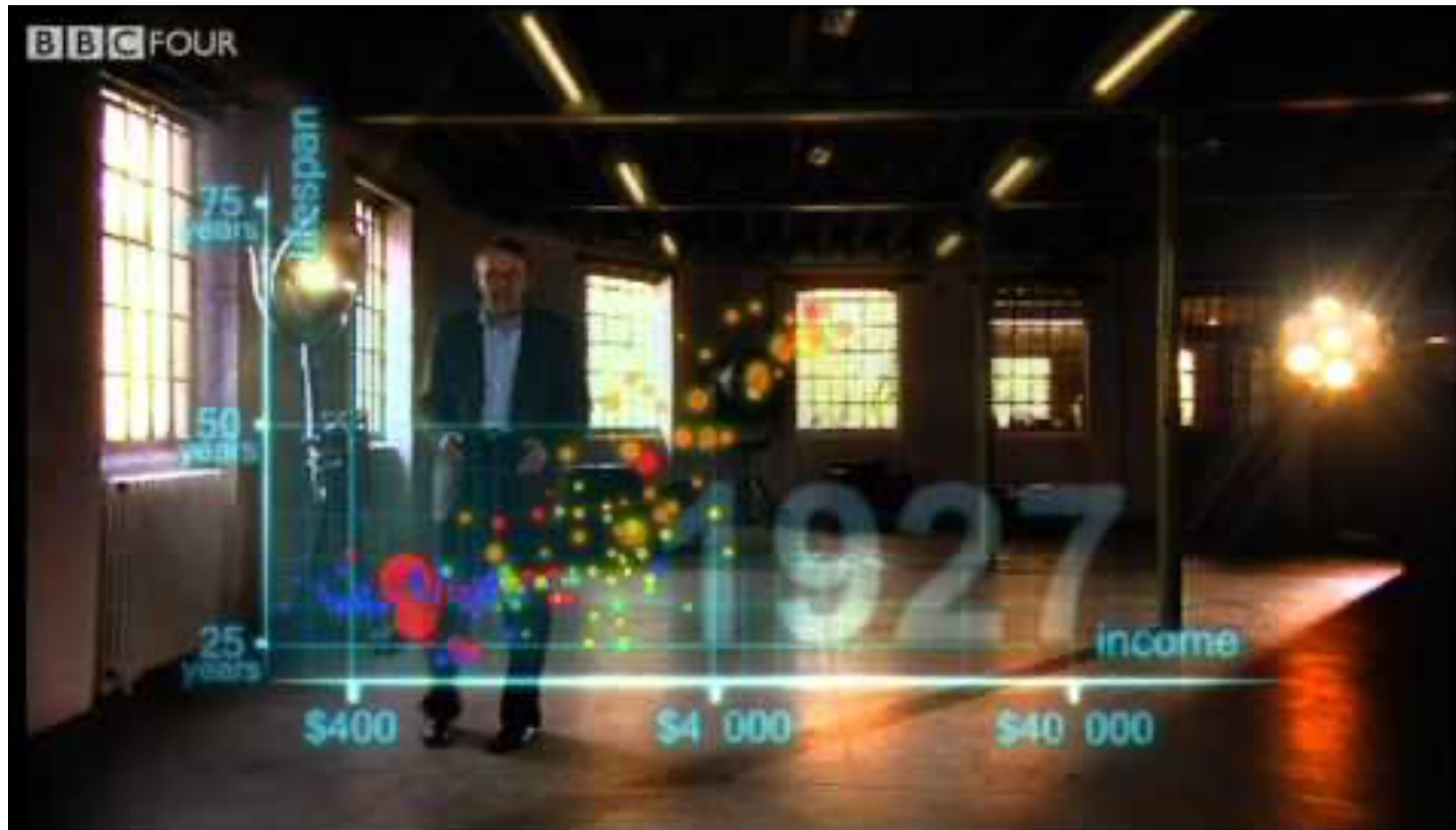
one country in 2007 (filtered)

Form

scatter, log x-axis



Hans Rosling's 200 Countries, 200 Years, 4 Minutes - The Joy of Stats - BB



Let's look at the fast-food data

You ask, we sort, we sketch together

The data: fastfood.csv

515 menu items · one row = one item · you've seen it in Labs 2 & 4

Columns

Step 0: name every type

C restaurant: 8 chains

C item: menu item name

Q calories, cal_fat

Q total_fat, sat_fat, trans_fat

Q cholesterol, sodium (mg)

Q total_carb, fiber, sugar, protein (g)

Q vit_a, vit_c, calcium (% daily value)

restaurant	item	calories	sodium
Mcdonalds	Artisan Grilled Chicken Sandwich	380	1110
Sonic	Garlic Parmesan Dunked Ultimate Chicken...	1350	...
Subway	Veggie Delite	...	...
...	...	...	...

With a neighbor · 2 min

Write down as many questions as you can.

Don't filter. Weird ones count.

Then we'll go around the room

Call them out and we'll sort

Sorting your questions

Q

Which item has the highest cholesterol?
Which of McD items has the highest sodium?

C + Q

How many items does McDonalds sell?

Highest and lowest average calories per restaurant?

Which restaurant has the highest average sodium?

Which restaurant offers the widest range in calories?

Is subway lower sodium on average than other restaurants?

Q + Q

Which item has the highest protein to calorie ratio?

How does cholesterol relate to total fat?

Q+Q+Q

How does total carbs, protein, and fiber relate to total calories across restaurants?

How does the variety of C, F and P compare across restaurants?

C + C + Q

Do the same items appear across restaurants and do the calories differ?

Let's run the recipe together

Question

Columns & types

One mark =

Form

Which chain's items are typically highest in calories?

Columns

C restaurant

Q calories

One mark =

one restaurant's items (a box)

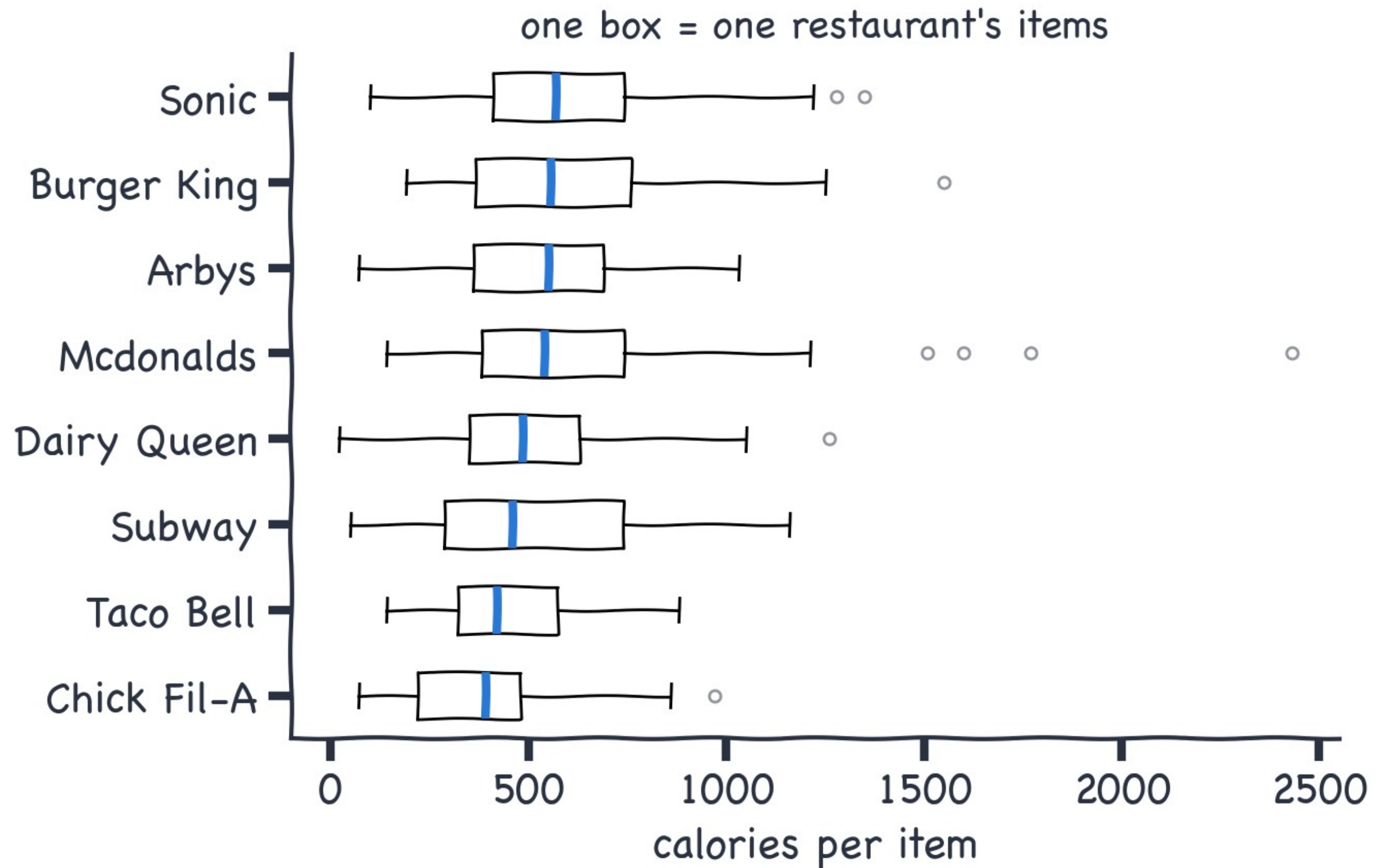
Form

box per restaurant, sorted by median

Why not...?

bar of means: hides the spread

bar of maxes: gets single highest point which answers the question but is very sensitive to menu changes, etc.



Are high-calorie items also high-sodium?

Columns

Q calories

Q sodium

One mark =

one menu item (a row)

Form

scatter



What share of an item's calories comes from fat?

Columns

Q $\text{cal_fat} \div \text{calories}$

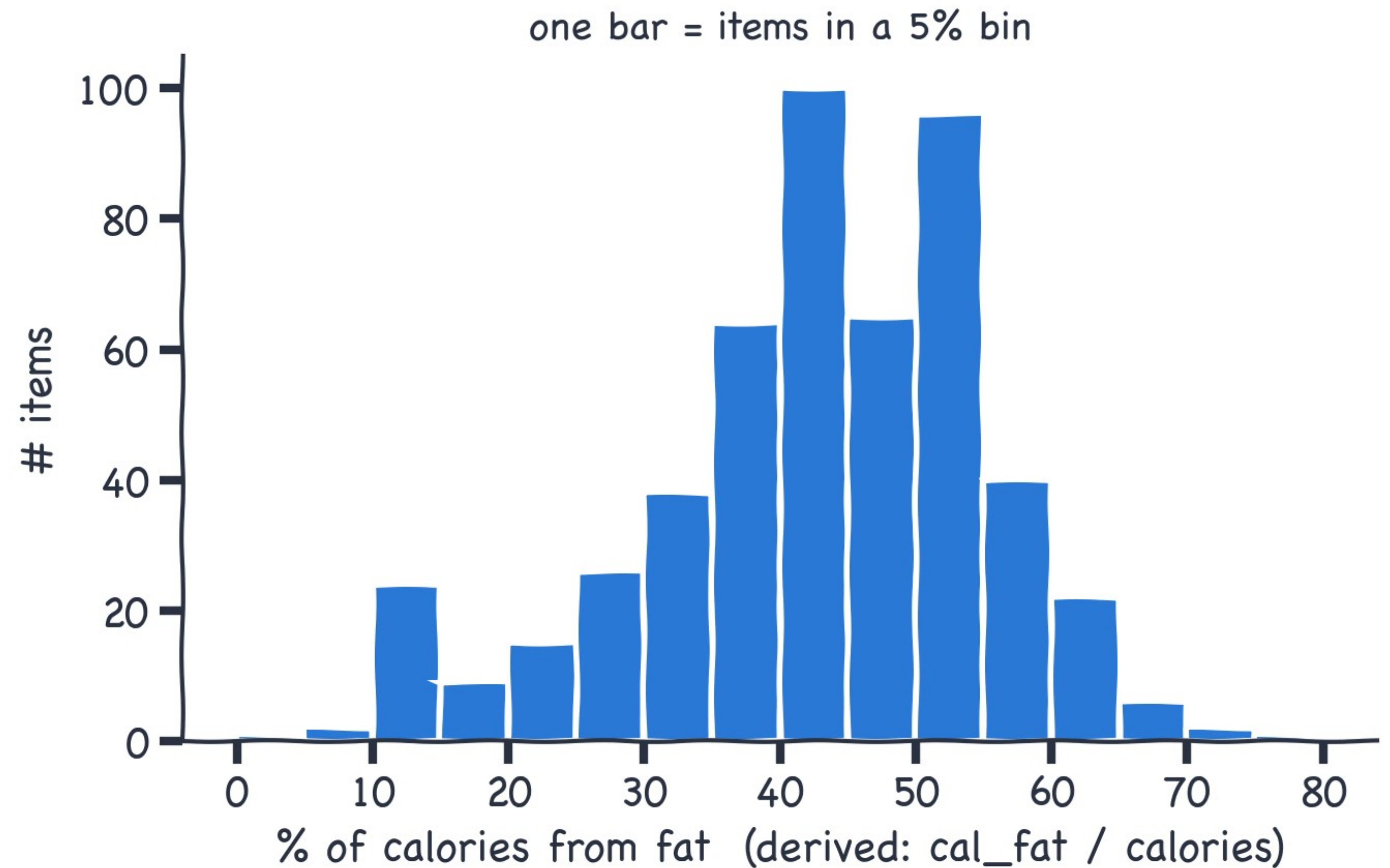
(derived, not in the table!)

One mark =

items in a 5% bin

Form

histogram



Your turn

Groups of 3–4 · pick a dataset card · paper + pens

Your turn (17 min)

Time	Step	What you do
5 min	Question storm	Write every question you can. Aim for 15+. Don't filter.
2 min	Tag	Label each with its guide row. Star 2 from different rows.
7 min	Sketch	Draw a plot for each starred question. Label the axes, write “one mark = ____”, and write the answer you expect.
3 min	Swap	Trade sketches with another group. Can they guess the question behind each one?

Pick a dataset

Spotify

28,356 songs

N track_name, artist
Q popularity (0–100)
T release_year (1957–2020)
N genre (6), subgenre (24)
Q danceability, energy, valence
Q acousticness, speechiness...
Q tempo, loudness, duration

Pokémon

949 Pokémon, Gen 1–7

O generation (1–7)
N type_1, type_2 (18 types)
Q hp, attack, defense
Q special_attack, special_defense
Q speed
Q height_m, weight_kg
Q base_experience

UFO sightings

80,332 reports, 1906–2014

T date_time
L latitude, longitude
L city, state, country
N ufo_shape (29)
Q duration_sec

Himalayas

76,519 climbers, 1905–2019

N peak_name (390 peaks)
T year
N season
N sex, citizenship
Q age
N role, hired (y/n)
N oxygen_used (y/n)
Q highpoint_metres
N success, died (y/n)

You just invented a taxonomy

Researchers have names for the questions you asked (Amar et al., 2005)

You asked...	Researchers call it
“What's the saltiest item?”	Find extremum
“Which chain has the highest typical calories?”	Compute derived value + sort
“Which items have $> 1,150$ mg sodium?”	Filter
“What's a typical flipper length?”	Characterize distribution
“Is GDP related to life expectancy?”	Correlate
“Which diamond is weirdly cheap for its size?”	Find anomalies
“Are there groups of similar songs?”	Cluster

The same guide, in researcher terms

Task	Sounds like...	Try
Retrieve value	What is the value of X for this case?	table · direct label on the mark
Filter	Which cases meet this condition?	highlight the matches, gray the rest
Compute derived value	What's the average / total / count / ratio?	compute first → bar of the summary, or one big number
Find extremum	Which is the highest / lowest?	sorted bar, top / bottom highlighted
Sort	Rank these.	sorted bar
Determine range	From what to what?	box · dot plot · min–max range bar
Characterize distribution	What's typical? How spread out?	histogram · box · strip
Find anomalies	What doesn't fit?	scatter or box, outliers labeled
Cluster	Are there groups of similar cases?	scatter · heatmap
Correlate	Does X go with Y?	scatter (Q+Q) · line (T+Q) · heatmap (C+C)

At home...

For your **Project 1** questions:

1. If your initial questions are too vague, think of ways to **sharpen** or **stretch** them first.
2. Then write the **attribute types** needed to answer each question
2. Based on the guide table, write out **possible chart types** you could use to answer each question
3. **Sketch one** of them. Label the axes and write “one mark = ____”.

YOU WILL TURN THIS IN ON WEDNESDAY IN LIEU OF A QUIZ.

Wednesday: fixing bad charts

Lots of bad charts are bad because the form doesn't match the question. Bring today's guide.